

Title Page

Engineering with Computers – Original Article

A GA-Warm-Started Mixed-Integer Nonlinear Framework for Discrete IS 800-Compliant Space-Truss Optimization

Short title: A GA-warm-started MINLP framework for code-constrained truss optimization

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Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The benchmark definitions, section catalogue data, optimization results, and post-processing files required to reproduce the tables and figures will be made available in a public repository at: [repository DOI/URL to be added before submission].

Code availability. The Python implementation used for the direct-stiffness analysis, IS 800:2007 member checks, optimization algorithms, statistical runs, and figure/table generation will be made available in the same repository at: [repository DOI/URL to be added before submission].

Use of AI-assisted technologies. During the preparation of this manuscript, the authors used Claude (Anthropic) and ChatGPT (OpenAI) to assist with L^AT_EX formatting, language editing, manuscript organisation, and journal-specific wording. The authors reviewed and edited all AI-assisted content and take full responsibility for the scientific accuracy and integrity of the work.